Statistical Thinking for Data Science

Week 4: Describing one variable honestly

Centre, spread, shape, and what an average hides

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Learning Objectives

By the end of this week you should be able to:

- read a distribution from a picture, and say what the picture is hiding;
- compute and choose between mean, median and trimmed mean, and say what each one minimises;
- compute variance, standard deviation, interquartile range and the median absolute deviation, and explain why there is more than one measure of spread;
- explain what makes a summary *robust*, and give the breakdown point of the common ones;
- apply the $1.5 \times$ IQR rule, and state how often it flags ordinary observations;
- standardise a value, and say when a coefficient of variation is meaningful;
- report a variable so that a reader can picture it without seeing the data.

4.1 Start with the distribution, not the average

The single worst habit in applied work is to meet a new variable by computing its mean. The mean is one number taken from a thing with a shape, and the shape is where the information is: whether the values cluster or spread, whether they are symmetric or trail off to one side, whether there are two groups hiding inside one column, whether some values are impossible.

Definition 4.1 (Distribution)**.** The pattern of values a variable takes: which values occur, and how often.

Throughout this chapter we use one small dataset, the durations in minutes of eleven unplanned stoppages in a processing plant:

$$4,\ 5,\ 6,\ 7,\ 8,\ 9,\ 11,\ 12,\ 14,\ 18,\ 46.$$

Small enough to compute by hand, and typical in shape: most stoppages are short, one is not.

4.2 Pictures of one variable

4.2.1 The histogram, and the choice nobody tells you about

A histogram groups values into bins and draws the counts. The bin width is yours to choose, and it changes the story.

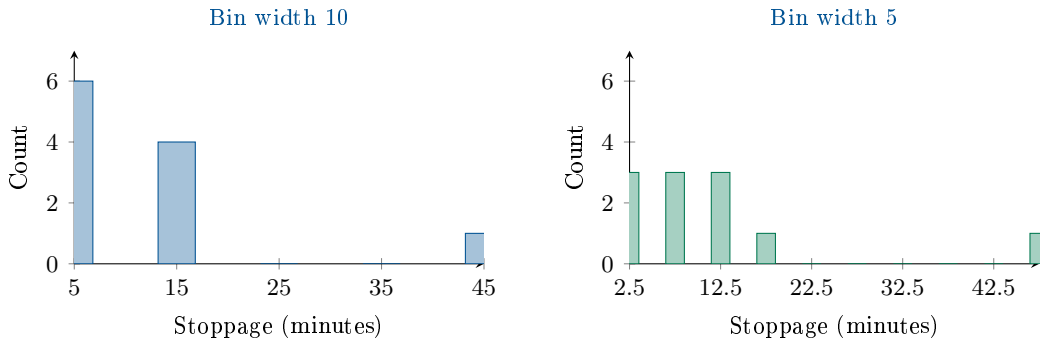


Figure 4.1: The same eleven stoppages. Wide bins suggest a single clump and an oddity; narrow bins show the short stoppages spread fairly evenly. Neither is wrong, and you should look at both before deciding what to say.

Always draw a histogram at two or three bin widths. A feature that survives all of them is probably real; one that appears at a single width usually is not.

4.2.2 Box plots, and what they conceal

A box plot draws the median, the quartiles and the fences. It is compact, excellent for comparing many groups at once, and blind to shape: a bimodal distribution and a flat one can produce identical boxes. Use it for comparison, never as your first look at a variable.

4.2.3 The empirical cumulative distribution

The ECDF plots, for each value x, the proportion of observations at or below x. It loses nothing, needs no bin choice, and answers questions of the form "what fraction of stoppages ran over 15 minutes?" directly. It is underused by students and heavily used by people who do this for a living.

4.3 Centre

Definition 4.2 (Mean, median, mode)**.** The **mean** is $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$. The **median** is the middle value in order. The **mode** is the most frequent value.

4.3.1 What each one minimises

This is the clearest way to see the difference, and it explains everything about when to use which.

- The mean minimises the sum of *squared* distances, $\sum_i (x_i - c)^2$. Differentiating with respect to c and setting the derivative to zero gives $-2\sum_i (x_i - c) = 0$, hence $c = \bar{x}$.
- The median minimises the sum of *absolute* distances, $\sum_i |x_i - c|$. Moving c up by a small amount changes the total by (number of points below c) minus (number above), which is zero exactly when c sits in the middle.

Squared distance punishes a far-away point enormously; absolute distance does not. That single fact is why one far-away point drags the mean and leaves the median where it was.

Example 1: the eleven stoppages, by hand

With $\sum x_i = 140$ and $n = 11$:

$$\bar{x} = \frac{140}{11} = 12.73 \text{ minutes}, \qquad \text{median} = 9 \text{ minutes}.$$

Solution

Eight of the eleven stoppages are shorter than the mean. A maintenance report quoting "average stoppage 12.7 minutes" describes no stoppage that actually happened: the figure sits in the empty gap between the ordinary stoppages and the one long one.

Drop the 46 and recompute: the mean falls to 9.40 and the median only to 8.5. One observation in eleven moved the mean by a quarter.

Report the median for anything skewed, which in engineering and economics means durations, incomes, costs, concentrations and waiting times. Report the mean when you need totals: mean stoppage times the number of stoppages is lost production, and the median will not do that job.

4.4 Spread

A centre without a spread is half a summary.

4.4.1 Variance and standard deviation

Definition 4.3 (Sample variance and standard deviation)**.**

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad s = \sqrt{s^2}.$$

For hand computation the algebraically equivalent form is easier:

$$s^2 = \frac{\sum_i x_i^2 - n\bar{x}^2}{n-1}.$$

Example 2: variance two ways

For the stoppages, $\sum x_i = 140$, $\sum x_i^2 = 3172$, $n = 11$, $\bar{x} = 12.727$.

$$s^2 = \frac{3172 - 11(12.727)^2}{10} = \frac{3172 - 1781.8}{10} = 139.02, \qquad s = 11.79 \text{ minutes}.$$

Solution

Note that $s = 11.79$ is almost as large as the mean of 12.73. When the standard deviation approaches the mean, the distribution is strongly skewed or heavy-tailed, and the pair "mean and SD" is a poor description. Dropping the 46 gives $s = 4.38$, a third of the size.

4.4.2 Why $n - 1$

Dividing by n gives a variance that is systematically too small, because the deviations are measured from $\bar{x}$, which was itself computed to sit as close to the data as possible. The data is nearer to its own mean than to the true mean. Dividing by $n - 1$ corrects the average size of that error exactly, making s^2 unbiased for the population variance. With $n = 11$ the correction is 10%; with $n = 200$ it is half a percent, which is why nobody argues about it in large samples.

4.4.3 Quartiles, IQR, and the median absolute deviation

Definition 4.4 (IQR and MAD)**.** IQR $= Q_3 - Q_1$, the range of the middle half. MAD $=$ median$_i\,|x_i - \text{median}(x)|$, usually scaled by 1.4826 so that it estimates the same thing as s for normal data.

Example 3: robust spread for the stoppages

Ordered, the data gives $Q_1 = 6.5$ and $Q_3 = 13$, so IQR $= 6.5$ minutes. The absolute deviations from the median of 9 are $5, 4, 3, 2, 1, 0, 2, 3, 5, 9, 37$, whose median is 3, so MAD $= 3$ and the scaled version is $1.4826 \times 3 = 4.45$.

Solution

Three measures of spread for one dataset: $s = 11.79$, IQR $= 6.5$, scaled MAD $= 4.45$. They disagree because they answer different questions. The standard deviation includes the 46 at full weight; the other two barely notice it. Quote s when the extremes matter to the decision, such as sizing a buffer; quote the IQR or MAD when describing the typical case.

4.5 Shape

Shape is the part of a distribution that centre and spread cannot express. Two features matter in practice.

Skew is asymmetry. A right-skewed variable has a long tail of large values, which is the normal state of affairs for durations, costs and concentrations. A quick measure is Pearson's coefficient,

$$\text{skew} \approx \frac{3(\bar{x} - \text{median})}{s},$$

which for the stoppages gives $3(12.727 - 9)/11.791 = 0.95$. The moment-based measure used by software is $g_1 = 2.25$ here. Both say the same thing qualitatively: the tail is on the right. Neither needs to be quoted in a report; the difference between mean and median already tells the reader.

Modality is how many clumps there are. One clump is ordinary. Two usually means two populations have been mixed, and the right response is to find the variable that separates them, not to describe the mixture.

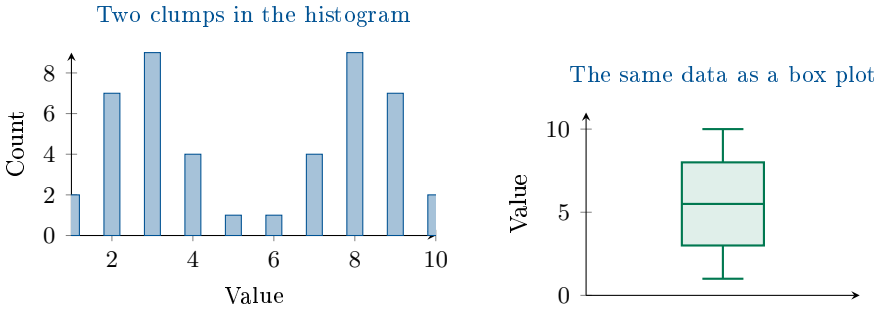


Figure 4.2: The box plot is a competent summary of a distribution that does not exist. Its median sits in the gap between the two clumps, where no observation falls.

4.5.1 Transforming a skewed variable

Taking logarithms pulls in a right tail, and for quantities that vary multiplicatively it is often the natural scale to work on.

Example 5: the stoppages on a log scale

The natural logarithms of the eleven durations are

$$1.39,\ 1.61,\ 1.79,\ 1.95,\ 2.08,\ 2.20,\ 2.40,\ 2.48,\ 2.64,\ 2.89,\ 3.83.$$

Their mean is 2.296 and their median is $\ln 9 = 2.197$, so Pearson's skew falls from 0.95 to 0.44: still right-skewed, but far closer to symmetric.

Solution

Exponentiating the mean of the logs gives the *geometric mean*,

$$\exp(2.296) = 9.93 \text{ minutes},$$

which sits near the median of 9 rather than near the arithmetic mean of 12.7. The geometric mean is the right average for quantities that compound or vary by factors: growth rates, concentrations, ratios.

The catch

Two of them. Logarithms are undefined at zero, so counts with zeros need care, and a model fitted on the log scale answers questions about the log scale: back-transforming a mean gives a median, not a mean. Say which scale a reported number lives on.

4.6 Robustness and the breakdown point

Definition 4.5 (Breakdown point)**.** The smallest fraction of the data that, if replaced by arbitrary values, can move a statistic arbitrarily far.

Statistic	Breakdown point	In words
Mean	$1/n \to 0$	One bad value ruins it
10% trimmed mean	0.10	Survives up to a tenth
Median	0.50	Survives up to half
Standard deviation	$1/n \to 0$	One bad value ruins it
IQR	0.25	Survives a quarter at each end
MAD	0.50	Survives up to half

Example 4: the trimmed mean as a compromise

Trim the largest and smallest values from the eleven stoppages and average the other nine:

$$\frac{5+6+7+8+9+11+12+14+18}{9} = \frac{90}{9} = 10.0 \text{ minutes.}$$

Solution

The trimmed mean, 10.0, sits between the median (9) and the mean (12.73). It uses every ordinary observation, unlike the median, while refusing to be dominated by the extreme, unlike the mean. State the trimming fraction whenever you report one, and decide it before you see the data, otherwise you are simply deleting inconvenient numbers with extra steps.

4.7 Outliers

An outlier is an observation far from the others. That is a statement about the data, not about correctness: the 46-minute stoppage is probably the most important row in the table.

The common rule flags anything outside

$$[\,Q_1 - 1.5\,\mathrm{IQR},\ Q_3 + 1.5\,\mathrm{IQR}\,].$$

For the stoppages that is $[-3.25,\ 22.75]$, so only the 46 is flagged.

Example 6: how often does the rule cry wolf?

Suppose the data really are normal, with no contamination at all. What fraction does the $1.5 \times$ IQR rule flag?

Solution

For a normal distribution the quartiles sit at ± 0.6745 standard deviations, so IQR $= 1.349$ and the fences fall at

$$\pm\,(0.6745 + 1.5 \times 1.349) = \pm 2.698 \text{ standard deviations.}$$

The probability of exceeding that in either direction is 0.698%, roughly one observation in 143. So in a clean sample of 500 normal observations, expect about three or four "outliers" that are nothing of the kind. The rule is a device for directing attention, not a test, and in a large dataset it will always find something.

What to do with an outlier, in order: **check it** against the source; **explain it** if you can; **report with and without it** if it changes the conclusion. Deleting it silently is misconduct, and it is also where the interesting engineering usually was.

4.8 Standardising, and comparing spreads fairly

The z-score expresses a value in standard deviations from the mean:

$$z = \frac{x - \bar{x}}{s}.$$

It makes quantities in different units comparable. A motor running at 78 °C when the fleet averages 70 with $s = 6$ has $z = 1.33$: warm, not alarming.

For comparing the *variability* of two things measured on different scales, use the coefficient of variation,

$$\text{CV} = \frac{s}{\bar{x}} \times 100\%.$$

Example 7: which engine is more consistent?

Engine A runs at $\bar{x} = 200$ °C with $s = 40$. Engine B runs at 400 °C with the same $s = 40$.

Solution

$\text{CV}_A = 40/200 = 20\%$ and $\text{CV}_B = 40/400 = 10\%$. Identical standard deviations, but relative to what each engine is doing, B is twice as steady.

The restriction

The CV only means something on a ratio scale, where zero means none. Computing a CV of temperatures in Celsius is a mistake, because 0 °C is an arbitrary point: the same two engines in kelvin give $\text{CV}_A = 40/473 = 8.5\%$ and $\text{CV}_B = 40/673 = 5.9\%$, different numbers for the same physical facts. Use the CV for mass, time, cost, concentration; not for Celsius, calendar years, or test scores.

4.9 Missing values

Before describing anything, count the missing. Three questions, in order: how many are missing, does missingness depend on the value itself (a sensor that fails when it overheats), and what did the software silently do with them. A mean computed over whatever was present is a mean over a self-selected subset, which is Week 3's problem wearing a different hat.

4.10 How to report a variable

One sentence, five ingredients: the number of observations, the centre, the spread, the shape, and anything unusual. For our stoppages:

> "Across 11 unplanned stoppages the median duration was 9 minutes (IQR 6.5), with a strongly right-skewed distribution: eight stoppages were under 12 minutes and one ran to 46 minutes, which accounted for a third of all lost time."

Note what that does. It uses the robust pair for the typical case, states the shape rather than implying symmetry, and reports the extreme as a fact about the plant rather than hiding it inside an average. Round sensibly too: 12.727272 minutes claims a precision that stopwatch data does not have.

4.11 Doing it on a computer

R

```
x <- c(4,5,6,7,8,9,11,12,14,18,46)
```

```
summary(x)                      # min, quartiles, median, mean, max
sd(x); IQR(x); mad(x)           # mad() is already scaled by 1.4826
mean(x, trim = 0.1)             # 10% trimmed mean
hist(x, breaks = 5); boxplot(x); plot(ecdf(x))
```

Python

```
import numpy as np, pandas as pd
x = pd.Series([4,5,6,7,8,9,11,12,14,18,46])

x.describe()                    # count, mean, sd, quartiles
x.median(); x.std(); x.quantile(.75) - x.quantile(.25)
(x - x.median()).abs().median() * 1.4826      # scaled MAD
from scipy import stats; stats.trim_mean(x, 0.1)
```

4.12 A checklist before you summarise

1. How many observations, and how many missing?
2. Plot it. Histogram at two bin widths, then an ECDF.
3. Are all values possible? Negative durations, ages of 200, zeros meaning "not recorded".
4. Is it one group or two? A bimodal histogram usually means a variable you forgot to split on.
5. Symmetric or skewed? That decides mean or median.
6. Any extreme values, and what do they turn out to be?
7. Have you rounded to a precision the measurement supports?

4.13 Summary

Week 4 in six lines

- Look at the distribution before computing anything.
- The mean minimises squared distance, the median absolute distance; skew decides which to report.
- Spread needs its own measure, and the robust ones disagree with s for a reason.
- Shape is the part centre and spread cannot say: note skew, and distrust one box plot of two clumps.
- Breakdown point tells you how much bad data a summary survives.
- The 1.5 IQR rule flags 0.7% of perfectly clean normal data.
- Report n, centre, spread, shape and the unusual, rounded honestly.

Exercises

1. For the data $3, 4, 4, 5, 7, 8, 9, 30$: compute the mean, median, Q_1, Q_3, IQR and the standard deviation.
2. For the same data, which values does the $1.5 \times$IQR rule flag? Recompute the mean and median with those values removed and comment.
3. A lecturer reports "mean mark 58, standard deviation 24" for a class of 120. Give two distributions consistent with those numbers that would call for different teaching responses.

4. Prove that the mean minimises $\sum_i (x_i - c)^2$ by differentiation, and state why the same argument fails for $\sum_i |x_i - c|$.

5. A dataset of 400 normal observations is screened with the 1.5 IQR rule. About how many points do you expect it to flag if nothing is wrong?

6. Monthly rainfall at a station has mean 120 mm and $s = 95$ mm. What does the ratio of those two numbers suggest about the shape, and which summary would you report to a civil engineer sizing a culvert?

7. Give the breakdown point of: the maximum; the midrange $(\min + \max)/2$; the 25% trimmed mean; the median.

8. Two boreholes give yields with $\bar{x}_1 = 2.4$ L/s, $s_1 = 0.6$ and $\bar{x}_2 = 9.0$ L/s, $s_2 = 1.8$. Compare their variability, and say which measure you used and why it is legitimate here.

9. A student standardises examination marks and reports a z-score of -2.6. Explain in one sentence to a non-statistician what that means.

10. A sensor records temperature every minute but fails whenever the reading exceeds 95 °C. Describe the effect on the reported mean, median and maximum, and say which is least damaged.

11. Using the class wellbeing data when it is available, describe one subscale following Section 4.9's five ingredients, in a single sentence.

12. Take any variable from a dataset you have, and produce: a histogram at two bin widths, an ECDF, and a one-sentence description. Bring all three to class.

13. A variable has mean 340, median 210 and $s = 290$. Compute Pearson's skew, say what shape it implies, and state which average you would report.

14. Three concentrations are measured as 2, 8 and 32 mg/L. Compute the arithmetic and geometric means, and explain which is the better summary of a quantity that changes by factors.

Answers to the odd-numbered exercises

1. $n = 8$, $\sum x = 70$, so $\bar{x} = 8.75$; median $= (5 + 7)/2 = 6$. Splitting at the median, the lower half is $3, 4, 4, 5$ and the upper half is $7, 8, 9, 30$, so $Q_1 = 4$, $Q_3 = 8.5$ and IQR $= 4.5$. The standard deviation is $s = 8.84$. (Software may report $Q_3 = 8.25$: there are several quartile conventions, and they differ in small samples. Say which you used, and never compare quartiles computed by different rules.)

3. For instance, a single unimodal clump around 58 (ordinary class, teach normally), or two clumps near 35 and 75 (two populations, perhaps students with and without the prerequisite, which calls for splitting the class rather than lecturing at the average).

5. About $400 \times 0.00698 \approx 2.8$, so two or three. Finding three flagged points in clean data is expected, not evidence of a problem.

7. Maximum: $1/n \to 0$. Midrange: $1/n \to 0$, since it depends on both extremes. 25% trimmed mean: 0.25. Median: 0.50.

9. The student scored about two and a half standard deviations below the class average, which, if marks are roughly bell-shaped, puts them in the lowest one percent or so of the class.

11. A model answer once the data exists, for example: "Among 214 responses the median depression score was 10 (IQR 12), right-skewed, with a quarter of students above 18 and a small group at the top of the scale."

13. $3(340 - 210)/290 = 1.34$, strongly right-skewed, so report the median of 210 and say that the mean is pulled up by a tail.

Before next week

Read *OpenIntro Statistics* Sections 2.1 to 2.2, and bring the three outputs from Exercise 12. Week 5 moves from one variable to two: how to describe a relationship, and why a relationship between two variables so often turns out to be about a third.